

Lecture 6: Set Notation, Defining Mathematics Using Sets, and Induction

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1 Abstract

In this lecture, we finished up our introduction to Sets and their notation. Using the formalized notation, we went into Russell's paradox, proving why not all statements in mathematics can be contained within sets.

We also introduced induction. Induction allows us to understand how a hypothesis is true for all possible inputs. By understanding the relationship between each element in a discrete chain, we can make conclusions about series and algorithms.

Finally, we introduced an algorithm which will require induction to prove it works for any array.

2 Recap: Set Notation

2.1 Set-Builder Notation Definition with Examples

Sets are commonly expressed using the Set-Builder Notation. For example, take the phrase:

A set of elements where $10 < x < 100$

This can be translated using set builder notation as:

$$S = \{x \in \mathbb{N} \mid 10 < x < 100\}$$

Dissecting the notation:

- Curly Braces ($\{\}$): Defines that the result is a set.
- Variable (x): This expresses each element (can be anything; it is only a placeholder).
- Constraint ($\in$): Constrains the Variable and declares that it "belongs" to the Domain.
- Domain (D): Declares the possible values for the variable (this domain is all natural numbers, $\mathbb{N}$).
- ($\mid$, $:$): Means "such that" or "where".
- Condition: What follows is the condition that defines everything that should belong in the set ($10 < x < 100$).

The resulting set will be defined as S .

Take the statement: **Primes** < 30 meaning **All the primes less than 30**. This translates to:

$$S = \{x \in \mathbb{N} \mid x \text{ is Prime} \wedge x < 30\}$$

3 Russell's Paradox

3.1 The Problem

Can we define mathematics using sets? Mathematicians were trying to prove that all statements, and therefore, the entity of mathematics, could be expressed only using sets.

3.2 Partitioning the Universe of Sets

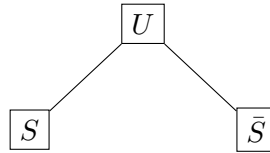
Let U be the universe of all sets. Consider partitioning U into two sets based on a chosen element, say the number 2:

- $S \rightarrow$ the set of all sets that have 2 as an element.
- $\bar{S} \rightarrow$ the set of all sets that do not have 2 as an element.

These two sets partition the whole space U . For example:

$$S = \{\{2\}, \{2, 3\}, \{2, 5, 7\}, \dots\}$$

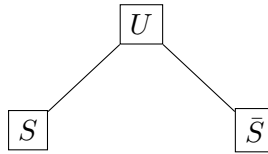
$$\bar{S} = \{\{\}, \{1, 3\}, \dots\}$$



This raises a natural question: **are there sets that satisfy any property we choose?** Consider the property " S contains itself":

$$S = \{x \in U \mid x \in x\}$$

$$\bar{S} = \{x \in U \mid x \notin x\}$$



sets that contain themselves sets that don't

3.3 The Paradox

Russell's Paradox: Let $\bar{S}$ be the set of all sets that do not contain themselves. We ask: does $\bar{S}$ contain itself?

Case 1: Suppose $\bar{S}$ contains itself. Then $\bar{S}$ belongs to S (the sets that contain themselves):

$$\bar{S} \in \bar{S} \implies \bar{S} \notin \bar{S} \implies \bar{S} \in S \cap \bar{S} \quad (\text{contradiction})$$

If it contains itself, it does not contain itself. It moves between two partitions, meaning that it would exist in two partitions of U at once, which is not possible.

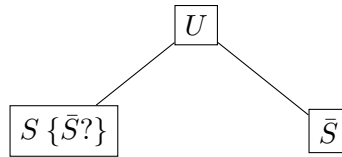
Formally, if $\bar{S} = \{x \mid x \notin x\}$, then:

$$\bar{S} \in \bar{S} \iff \bar{S} \notin \bar{S}$$

Case 2: Suppose $\bar{S}$ does not contain itself. Then it belongs to $\bar{S}$ (the container). But if it belongs to that set, it becomes a set that contains itself — moving between partitions and creating a contradiction:

$$\bar{S} \notin \bar{S} \implies \bar{S} \in \bar{S} \implies \bar{S} \notin \bar{S}$$

Either way, $\bar{S}$ cannot be consistently placed in either branch of the partition of U :



Since $\bar{S}$ is missing from $\bar{S}$, it should be in $\bar{S}$ — a contradiction. This is Russell's Paradox: it shows that not every property can be used to define a set, since doing so can produce a set that both must and must not contain itself.

4 Induction

4.1 The Principle of Induction

For a property that we want to hold for all n :

$$\forall n : \text{Property}(n) \text{ is true}$$

Proof by induction proves that something is true at the beginning (the base case). Then, we prove that $P(i) \implies P(i+1)$ is true. This can be applied for any i , at any point, proving that the next case is also true — i.e., proving that moving through the chain $0, 1, 2, \dots, i \rightarrow i+1, \dots, n$ preserves truth.

The three steps of an induction proof:

1. Show $P(0)$ (base case).
2. Assume $P(i)$ is true (inductive hypothesis).
3. Deduce $P(i+1)$ is true (inductive step).

A chain works for discrete math because there is always a "next" item, forming a chain. An example of Induction is found in Peano's Axioms. The fifth axiom says:

$$\text{If } K \text{ is a set where } 0 \in K \text{ and } (\forall n \in K \implies S(n) \in K), \text{ then } K = \mathbb{N}$$

Assuming that the base case ($0 \in K$). Using induction, you can prove $(\forall n \in K \implies S(n) \in K)$, by using n for the inductive hypothesis, and that the inductive step is true for the successor function. Therefore, it proves a single, unbroken chain of numbers that always starts from 0.

4.2 Example: Sum of the First n Natural Numbers

Prove $\forall n \in \mathbb{N}$:

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

1. Show $P(0)$ (base case).

$$P(0) : 0 = \frac{0 \cdot (0+1)}{2} = 0$$

2. Assume $P(i)$ is true (inductive hypothesis).

$$P(i) : 1 + 2 + 3 + \dots + i = \frac{i(i+1)}{2}$$

3. Deduce $P(i + 1)$ is true (inductive step).

$$\begin{aligned} P(i + 1) : 1 + 2 + 3 + \cdots + i + (i + 1) &= \frac{i(i + 1)}{2} + (i + 1) \\ &= (i + 1) \left[\frac{i}{2} + 1 \right] = \frac{(i + 1)(i + 2)}{2} = P(i + 1) \end{aligned}$$

Therefore, $P(i) \implies P(i + 1)$.

4.3 Direct Proof (Gauss's Pairing Trick)

As an alternative to induction, consider the direct proof of $1 + 2 + 3 + \cdots + 100$. Let's pair up the integers:

$$(1 + 100), (2 + 99), (3 + 98), \dots$$

Each pair sums to 101 ($n + 1$). Since we pairing up 100 integers (n), it will result in 50 pairs ($\frac{n}{2}$). This gives:

$$\sum_{n=1}^{100} n = \frac{n}{2}(n + 1)$$

4.4 Example: Sum of Squares

Prove $\forall n \in \mathbb{N}$:

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n + 1)(2n + 1)}{6}$$

1. Show $P(0)$ (base case).

$$P(0) = 0$$

2. Assume $P(i)$ is true (inductive hypothesis).

$$P(i) = 1^2 + 2^2 + \cdots + i^2 = \frac{i(i + 1)(2i + 1)}{6}$$

3. Deduce $P(i + 1)$ is true (inductive step).

$$P(i + 1) = \frac{i(i + 1)(2i + 1)}{6} + (i + 1)^2 = (i + 1) \left[\frac{i(2i + 1)}{6} + i + 1 \right]$$

Pattern matching: we want to show this equals $\frac{(i + 1)(i + 2)(2i + 3)}{6} = P(i + 1)$.

$$\frac{(i + 1)}{6} [2i^2 + i + 6i + 6] = \frac{(i + 1)}{6} [2i^2 + 7i + 6]$$

Factor by grouping: $2 \times 6 = 12$, $3 \times 4 = 12$, and $3 + 4 = 7$ (the sum of the middle coefficient), so:

$$\begin{aligned} \frac{(i + 1)}{6} [2i^2 + 4i + 3i + 6] &= \frac{(i + 1)}{6} [2i(i + 2) + 3(i + 2)] \\ &= \frac{(i + 1)(2i + 3)(i + 2)}{6} \end{aligned}$$

This matches $P(i + 1)$. Therefore, true by induction.

4.5 Example: Divisibility Proof

Prove $\forall n \in \mathbb{N}: 6^n - 1$ is divisible by 5.

1. Show $P(0)$ (base case).

$$P(0) : 6^0 - 1 = 1 - 1 = 0, \quad 5 \times 0 \text{ divisible}$$

2. Assume $P(i)$ is true (inductive hypothesis).

$$P(i) : 6^i - 1 = 5k, \quad k \in \mathbb{Z} \implies 6^i = 5k + 1$$

3. Deduce $P(i + 1)$ is true (inductive step).

$$P(i + 1) : 6^{i+1} = 5k + 1$$

$$6 \cdot 6^i = 6(5k + 1)$$

$$6^{i+1} = 30k + 6$$

$$6^{i+1} - 1 = 30k + 5 = 5(6k + 1)$$

Since $6k + 1 \in \mathbb{Z}$, $6^{i+1} - 1$ is divisible by 5. True by induction.

4.6 Example: The Postage Stamp Problem

Prove: You have 5-cent stamps and 8-cent stamps. Prove that $\forall k \geq 28$, you can generate exactly k cents of postage using only 5-cent and 8-cent stamps. That is, $k = 5a + 8b$ for some non-negative integers a, b .

1. Show $P(28)$ (base case).

$$P(28) : 28 = 4 \times 5 + 1 \times 8 \quad (a = 4, b = 1)$$

2. Assume $P(i)$ is true (inductive hypothesis), for some $i \geq 28$.

$$P(i) : i = 5a + 8b \quad \text{for some non-negative integers } a, b$$

3. Deduce $P(i + 1)$ is true (inductive step). We want to write $i + 1$ as $5a' + 8b'$ using the a, b from above.

Two exchange identities. Notice that trading stamps in these two ways changes the total by exactly +1:

- Trade 3 fives for 2 eights: $3 \times 5 = 15$, and $15 + 1 = 16 = 2 \times 8$.
- Trade 3 eights for 5 fives: $3 \times 8 = 24$, and $24 + 1 = 25 = 5 \times 5$.

Why at least one trade is always available. Using fewer than 3 of *each* denomination (i.e. $a \leq 2$ and $b \leq 2$) can make at most $2(5) + 2(8) = 26$ cents. Since $i \geq 28 > 26$, any representation of i must have $a \geq 3$ or $b \geq 3$ (or both) — so one of the two trades below is always possible.

Case 1: $a \geq 3$. Trade 3 fives for 2 eights:

$$i + 1 = 5(a - 3) + 8b + \underbrace{(3 \times 5 + 1)}_{=2 \times 8} = 5(a - 3) + 8(b + 2)$$

which is a valid representation since $a - 3 \geq 0$.

Case 2: $b \geq 3$. Trade 3 eights for 5 fives:

$$i + 1 = 5a + 8(b - 3) + \underbrace{(3 \times 8 + 1)}_{= 5 \times 5} = 5(a + 5) + 8(b - 3)$$

which is a valid representation since $b - 3 \geq 0$.

Since at least one of the two cases always applies, $i + 1$ can always be written as $5a' + 8b'$ for non-negative integers a', b' . Therefore $P(i) \implies P(i + 1)$, and by induction, $P(k)$ holds for all $k \geq 28$.

5 Next Lecture: Using Induction for Algorithms

5.1 The Algorithm

Induction is very useful for understanding and logically proving if an algorithm is useful.

Problem: Given an array, find the maximum continuous (contiguous) sum.

Input: $[-2, 1, -3, 4, -1, 2, 1, -5, 4]$

Output: Maximum continuous sum.

Most trivial approach:

```
for i = 0 -> n-1
  for j = 0 -> n-1
    sum(A[i..j])
```

This grows at n^3 , i.e. $O(n^3)$.

5.2 Next Lecture: How we will use Induction

Improving toward $O(n^2)$ using a running sum:

```
for i = 0 -> n-1
  prev = 0
  for j = 0 -> n-1
    prev += A[j]
```

Considerations for further improving to $O(n)$:

- Negative running sum $\rightarrow$ very bad.
- Small negative $\rightarrow$ tolerable.
- If a negative value removes/erases the running sum, then it is not valid.

We will use induction next lecture to prove that this algorithm works correctly for any array.

6 Questions while Writing Notes

1. For the Postage Stamp Problem, how do you increment the total by 1? Specifically, how do you transition from $28 \rightarrow 29 \rightarrow 30$, and when should you trade 5-cent stamps for 8-cent stamps?
2. Does $\bar{s}$ represent the container or the empty set?
3. What is the process for determining the $(i + 1)$ step in a proof where you cannot directly add 1?
4. Why is adding $(3 \times 5 + 1)$ necessary when assuming the statement must be true for the base case? How does the $(i + 1)$ step connect to the expression $5(a - 3)$?
5. For the divisibility proof, does the base definition of divisibility apply when the result is 0?